

National University of Computer and Emerging Sciences, Lahore Campus



Course: Analog and Digital Communications
Program: BS(Electrical Engineering)
Duration: 1 Hour
Paper Date: 2-Oct-18
Section: ALL
Exam: MID-I

Course Code: EE323
Semester: Fall 2018
Total Marks: 45
Weight: 20%
Page(s): 3
Roll No: 15-4521

Instruction/Notes: Attempt all parts of a question together.
Fourier Transforms Table is provided

[15]

Question # 1 [CLO: 1]

A low-pass filter transfer function $H(\omega)$ is given by

$$H(\omega) = \tau \left[\frac{\sin\left(\frac{\omega\tau}{2}\right)}{\frac{\omega\tau}{2}} \right] e^{-j5\omega}$$

- Is this a distortion-less system, if yes explain, if no then identify the type of distortion introduced by this system.
- A signal $g(t) = \delta(t)$ is applied at the input of this filter. Find the output $y(t)$ and describe the effect of channel on input signal.

[15]

Question # 2 [CLO: 2]

Given the signal

$$g(t) = e^{-t}u(t)$$

- Show that $g(t)$ is an energy signal.
- Find the autocorrelation function $\psi_g(\tau)$ of $g(t)$.
- Show how from $\psi_g(\tau)$ you can deduce the energy spectral density $\Psi_g(\omega)$ of $g(t)$.
- Find the ESD of output signal $\Psi_y(\omega)$ if the signal $g(t)$ is passed through a filter whose frequency response is $H(\omega) = e^{-j\omega T}$, where T is a constant.
- Find the output signal energy E_y using output energy spectral density $\Psi_y(\omega)$ found in part d.

[15]

Question # 3 [CLO: 3]

The spectrum of the message signal $m(t)$ is shown in Fig. 1 below.

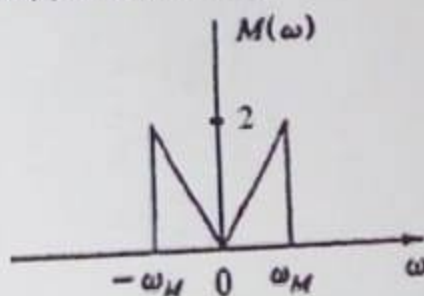


Figure 1 Spectrum of $m(t)$

- a) The signal $m(t)$ is applied to a system shown in Fig 2. Analyze the system and sketch the spectrum at points A and B . Properly label the axis information. Determine the bandwidth of signals at A and B .

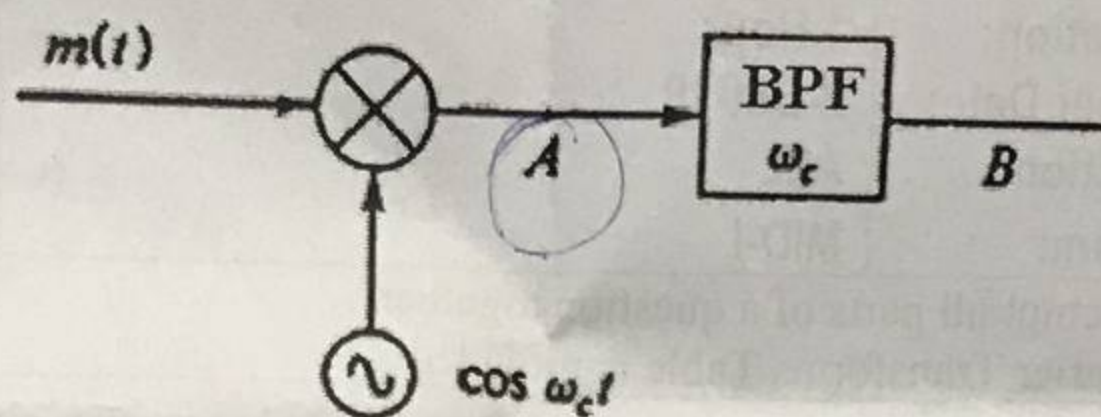


Figure 2

- b) The signal $m(t)$ is applied to a system shown in Fig 3. Analyze the system and sketch the spectrum of output $x(t)$. Properly label the axis information. Determine the bandwidth of signal $x(t)$.

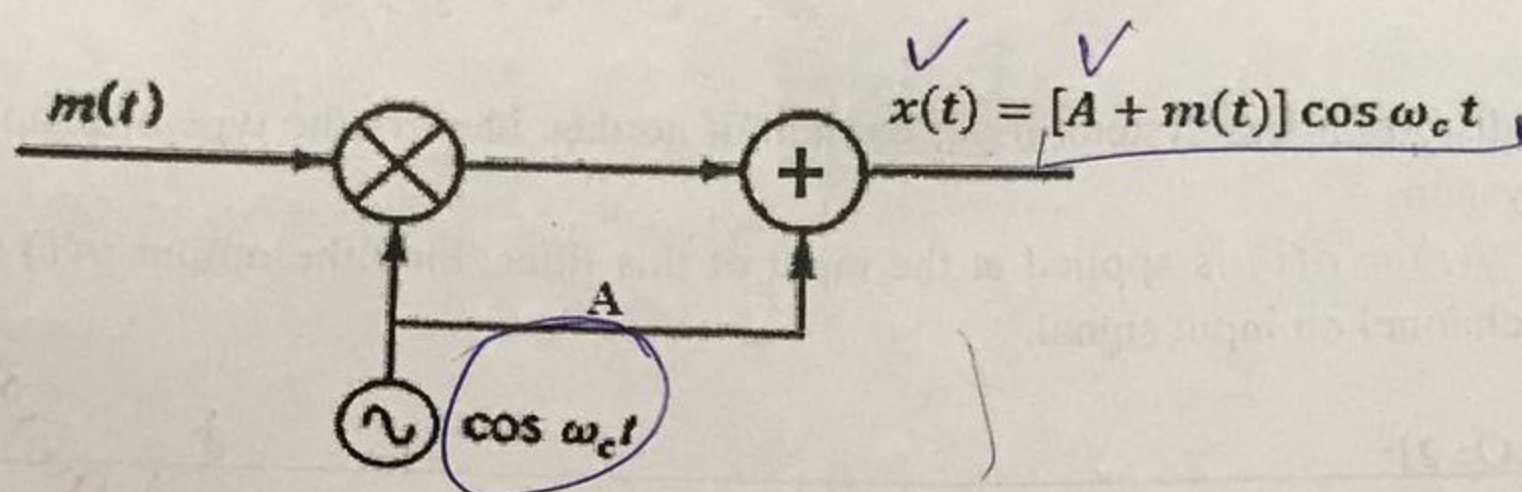


Figure 3

Note: $\omega_c \gg \omega_M$

Handwritten notes and sketches:

- $x(t) = [A + m(t)] \cos \omega_c t$
- $m(t) \cos \omega_c t$
- $\frac{1}{2} m(\omega + \omega_c) + \frac{1}{2} m(\omega - \omega_c)$
- $\frac{e^{-j\omega t}}{2}$
- $\frac{e^{j\omega t}}{2}$
- $\frac{1}{2} m(\omega + \omega_c)$
- $\frac{1}{2} m(\omega - \omega_c)$